

Real-Time Motorcycle Helmet Violation Detection in Developing Regions Using YOLO11 with Transfer Learning

Rukaiya Sultana Mim

ID: 2023100000643

Department of Computer Science and Engineering

Southeast University

Tejgaon, Dhaka, Bangladesh

Abstract—Motorcycle accidents are a leading cause of fatalities in developing countries, largely exacerbated by the failure to wear safety helmets. Manual enforcement of helmet laws by traffic police is labor-intensive, inconsistent, and prone to human error. In this paper, we propose an automated deep learning approach using the latest YOLO11 object detection architecture to detect motorcycle helmet violations. Utilizing the Mapua University Motorcycle Helmet Dataset, we applied transfer learning by fine-tuning a YOLO11-Small model pre-trained on the COCO dataset. Initially, the dataset comprised five highly overlapping classes. However, through strategic dataset simplification—merging visually similar helmet sub-categories into two distinct classes (“Helmet” and “No Helmet”)—we significantly reduced classification complexity. This approach yielded an overall mAP50 of 87.0%, with the critical “No Helmet” class achieving a remarkable 90.9% mAP50. Furthermore, the model operates at an inference speed of 4.6ms per image, demonstrating its high viability for real-time traffic surveillance and smart city infrastructure.

I. INTRODUCTION

Road safety remains a critical challenge in developing nations like Bangladesh, where motorcycles are a primary mode of transportation due to their affordability and maneuverability in heavy traffic. According to local traffic authorities, motorcycles account for a disproportionately high number of road fatalities, with head injuries being the leading cause of death. Despite strict traffic regulations mandating helmet use, compliance remains dangerously low. Traditional enforcement relies on traffic police manually monitoring intersections, which is labor-intensive, inconsistent, and often impossible on high-speed highways.

Recent advancements in Deep Convolutional Neural Networks (DCNNs) and computer vision have opened new avenues for automated traffic monitoring. For instance, research on Bangladeshi vehicle classification using transfer learning has proven that deep learning can effectively solve region-specific traffic challenges [1]. Building upon this concept, this paper applies the state-of-the-art YOLO11 architecture to automatically detect helmet violations from traffic camera feeds. Unlike traditional two-stage detectors, YOLO operates in a single pass, making it exceptionally fast and suitable for live video surveillance.

The primary contributions of this study are threefold: (1) We implement the newly released YOLO11-Small architecture for the task of helmet detection, leveraging transfer learning from the COCO dataset. (2) We provide an empirical analysis demonstrating how strategic dataset simplification—reducing overlapping sub-categories into binary target classes—drastically improves detection accuracy, boosting mAP50 from 55.8% to 87.0%. (3) We validate the model’s real-time capabilities on regional traffic data, proving its readiness for deployment in developing smart cities.

II. RELATED WORK

Object detection in traffic scenarios has been extensively researched, evolving from traditional Haar cascades to sophisticated deep learning models. Early approaches to helmet detection utilized frameworks like Faster R-CNN [4], which, while accurate, suffered from high computational latency, making them unsuitable for real-time video processing.

The introduction of the YOLO (You Only Look Once) paradigm revolutionized real-time detection. Subsequent iterations, such as YOLOv3, YOLOv5, and YOLOv8, have been frequently applied to helmet detection tasks. However, many existing studies attempt to classify helmets into granular sub-categories (e.g., half-face, full-face, invalid), which often leads to high misclassification rates due to the visual similarity between these classes from a distance.

Transfer learning has also become a cornerstone of traffic vision systems. As demonstrated by Farid et al. [1], leveraging pre-trained weights from large datasets like ImageNet or COCO allows models to generalize effectively on smaller, region-specific datasets, such as Bangladeshi vehicles. Our work extends this principle by applying the latest YOLO11 architecture with COCO pre-training to the specific problem of helmet compliance, while innovating on the class-strategy to optimize real-world enforcement accuracy.

III. METHODOLOGY

A. Dataset Analysis and Preprocessing

We utilized the Mapua University Motorcycle Helmet Detection Dataset, this dataset contains 1650 images, which features urban traffic scenarios highly representative of developing South Asian cities. Originally, the dataset comprised five classes: "Rider," "Full Face Helmet," "Half Face Helmet," "Invalid Helmet," and "No Helmet."

During preliminary experiments, training on the five-class schema yielded a sub-optimal overall mAP50 of only 55.8%. Upon analysis, we identified that the "Half Face," "Invalid," and "Full Face" classes share high visual overlap. For example, a half-face helmet viewed from a slight angle can easily be misclassified as an invalid or full-face helmet by the network.

For the purpose of automated law enforcement, the exact style of the helmet is secondary to whether the rider is protected. Therefore, we merged the three helmet categories into a single "Helmet" class and retained "No Helmet" as the second class. The "Rider" class was dropped as it does not contribute to violation logic. The class mapping is detailed in Table I.

TABLE I
CLASS MAPPING STRATEGY FOR DATASET SIMPLIFICATION

Original Class	Merged Class
Full Face Helmet Half Face Helmet Invalid Helmet	3*Helmet (0)
No Helmet	No Helmet (1)
Rider	Dropped

B. YOLO11 Architecture and Transfer Learning

YOLO11 is the latest iteration in the YOLO family, featuring an enhanced architecture with optimized CSP (Cross Stage Partial) bottlenecks and improved feature pyramid networks for better small-object detection. We employed the YOLO11-Small (YOLO11s) variant, which offers an optimal balance between parameter count and detection precision.

Instead of training the network from scratch, we applied **Transfer Learning**. We initialized the YOLO11 model with weights pre-trained on the Microsoft COCO dataset [3]. Because the COCO dataset already contains classes like "person" and "bicycle," the pre-trained weights provided a robust foundational feature extractor. We then fine-tuned the entire network on our two-class helmet dataset, allowing the model to adapt its generalized feature extraction capabilities to the specific nuances of helmet detection.

C. Training Configuration

The model was trained using the Ultralytics framework on an NVIDIA Tesla T4 GPU. The input image resolution was set to 640x640 pixels, with a batch size of 8 to accommodate GPU memory constraints. The Adam optimizer was utilized with an initial learning rate of 0.01. Additionally, the Ultralytics framework automatically applied Mosaic data augmentation

during training, which combines four images into one to help the model better detect small objects like helmets in crowded scenes. To prevent overfitting and ensure optimal generalization, an early stopping mechanism was implemented with a patience of 15 epochs. The training process automatically terminated at epoch 49, having identified the best-performing model weights at epoch 34.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Training Dynamics

Figure 1 illustrates the training and validation loss and metrics curves over the training period. The model converged rapidly within the first 10 epochs. The use of early stopping at epoch 49 (capturing the best weights from epoch 34) confirmed that the model reached its peak generalization capability without overfitting the training data.

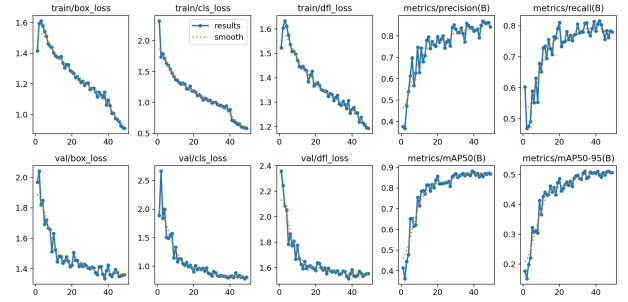


Fig. 1. Training and validation loss/metrics curves over epochs.

B. Detection Performance

The model's performance was evaluated using the Mean Average Precision (mAP) metric at an Intersection over Union (IoU) threshold of 0.5. The confusion matrix (Figure 2) demonstrates the model's strong ability to distinguish between riders with and without helmets following the class-merging strategy.

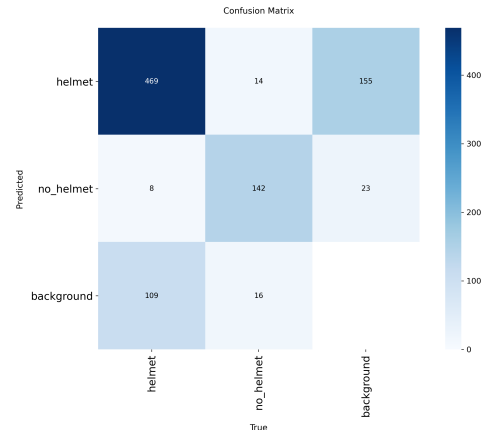


Fig. 2. Confusion matrix for the binary helmet detection model.

The final evaluation metrics are summarized in Table II. The simplified two-class model achieved an overall mAP50 of **87.0%**. Most notably, the "No Helmet" class achieved an exceptional mAP50 of **90.9%** with a precision of 86.5%, indicating high reliability in identifying violators with minimal false positives. This represents a massive improvement over the initial five-class experiment, which yielded an overall mAP50 of only 55.8% and struggled to separate "Invalid" and "Half Face" helmets.

TABLE II
FINAL EVALUATION METRICS PER CLASS

Class	Precision	Recall	mAP50	mAP50-95
Overall	81.3%	80.5%	87.0%	51.2%
Helmet	76.1%	79.4%	83.2%	49.2%
No Helmet	86.5%	81.7%	90.9%	53.1%

Figure 3 shows sample predictions where the model successfully draws bounding boxes around riders, accurately classifying helmet compliance even in dense and dynamic traffic scenarios.



Fig. 3. Sample detection results on validation data showing successful "Helmet" and "No Helmet" classification.

C. Real-Time Feasibility

Beyond accuracy, the feasibility of a traffic enforcement system depends on its processing speed. The YOLO11-Small model achieved an inference speed of 4.6ms per image. This translates to a processing capability of over 200 frames per second (FPS), which far exceeds the standard 30 FPS requirement for real-time video streams. This confirms that the model can be deployed on edge devices or traffic cameras without bottlenecking the video feed.

V. CONCLUSION

In this paper, we implemented a YOLO11-Small model with transfer learning to detect motorcycle helmet violations in developing regions. By strategically merging visually similar helmet sub-categories into a simplified binary structure ("Helmet" and "No Helmet"), the model's mAP50 jumped from 55.8% to 87.0%. The system achieved a 90.9% mAP50 on the critical "No Helmet" class, proving that lightweight, modern object detectors can be effectively and reliably deployed to enhance road safety. The inference speed of 4.6ms per

image further validates its suitability for real-time deployment. Future work could integrate this system with automatic license plate recognition (ALPR) to issue automated fines to traffic violators, creating a complete end-to-end enforcement system.

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